# COHEN-MACAULAYNESS OF A LOCAL RING, ITS HENSELIZATION AND ITS STRICT HENSELIZATION

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*Allowed standard prerequisites (editorial).*

- The henselization and strict henselization of a Noetherian local ring are Noetherian. Lemma \ref{lemma-henselization-noetherian}, cited at proof lines 11181-11182. Makes the Cohen-Macaulay condition meaningful for the henselizations.
- Depth is unchanged by henselization and strict henselization. Lemma \ref{lemma-henselization-depth}, cited at proof lines 11184-11187. Gives one of the two invariants whose equality defines Cohen-Macaulayness.
- Dimension is unchanged by henselization and strict henselization. Lemma \ref{lemma-henselization-dimension}, cited at proof lines 11186-11188. Gives the other invariant.

*Source reference.* Original lemma 06LM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8471–8471. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 06LJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8468–8468. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 06LL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8470–8470. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 06LK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8469–8469. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $R$ be a Noetherian local ring. The following are equivalent: $R$ is Cohen-Macaulay, the henselization $R^h$ of $R$ is Cohen-Macaulay, and the strict henselization $R^{sh}$ of $R$ is Cohen-Macaulay.*

*Proof.* By Lemma 06LJ we know that $R^h$ and $R^{sh}$ are Noetherian, hence the lemma makes sense. Since we have $\text{depth}(R) = \text{depth}(R^h) = \text{depth}(R^{sh})$ and $\dim(R) = \dim(R^h) = \dim(R^{sh})$ by Lemmas 06LL and 06LK we conclude. □